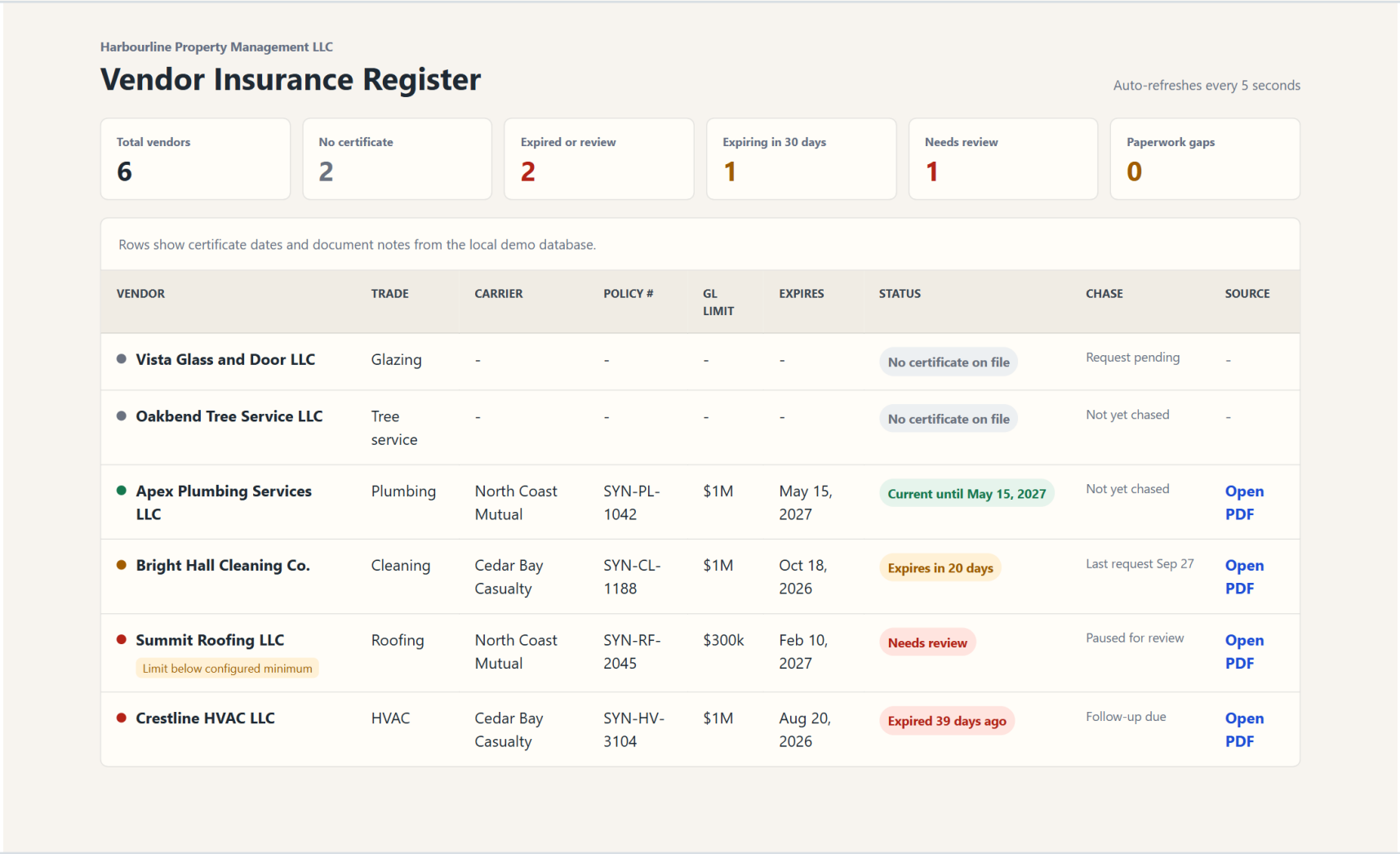
AUTOMATION CASE STUDY

Vendor COI chasing

Six n8n workflows connect certificate intake, validation, a vendor register, renewal chasing and human approval.



Harbourline Property Management LLC

Vendor Insurance Register

Auto-refreshes every 5 seconds

Total vendors	No certificate	Expired or review	Expiring in 30 days	Needs review	Paperwork gaps
6	2	2	1	1	0

Rows show certificate dates and document notes from the local demo database.

VENDOR	TRADE	CARRIER	POLICY #	GL LIMIT	EXPIRES	STATUS	CHASE	SOURCE
Vista Glass and Door LLC	Glazing	-	-	-	-	No certificate on file	Request pending	-
Oakbend Tree Service LLC	Tree service	-	-	-	-	No certificate on file	Not yet chased	-
Apex Plumbing Services LLC	Plumbing	North Coast Mutual	SYN-PL-1042	$1M	May 15, 2027	Current until May 15, 2027	Not yet chased	Open PDF
Bright Hall Cleaning Co.	Cleaning	Cedar Bay Casualty	SYN-CL-1188	$1M	Oct 18, 2026	Expires in 20 days	Last request Sep 27	Open PDF
Summit Roofing LLC Limit below configured minimum	Roofing	North Coast Mutual	SYN-RF-2045	$300k	Feb 10, 2027	Needs review	Paused for review	Open PDF
Crestline HVAC LLC	HVAC	Cedar Bay Casualty	SYN-HV-3104	$1M	Aug 20, 2026	Expired 39 days ago	Follow-up due	Open PDF

REGISTER OUTPUT / rendered by the project dashboard workflow

Illustrative synthetic rows

SYSTEM DESIGN

One connected automation

n8n handles orchestration. Postgres stores vendors, certificates, coverage, notices and outreach state. Original PDFs stay linked to the register.

01 **COI Intake**

Gmail, upload form or fixture -> PDF storage, extraction and validation

02 **Source Document Route**

Serves the stored original PDF behind a certificate link

03 **Daily Watch and Chase**

Selects due vendors, drafts requests and records pending outreach

04 **Vendor Approval Request**

One execution per vendor; sends only after a human approves

05 **Non-PDF Reply Notice**

Captures replies that require manual attention rather than PDF processing

06 **COI Dashboard**

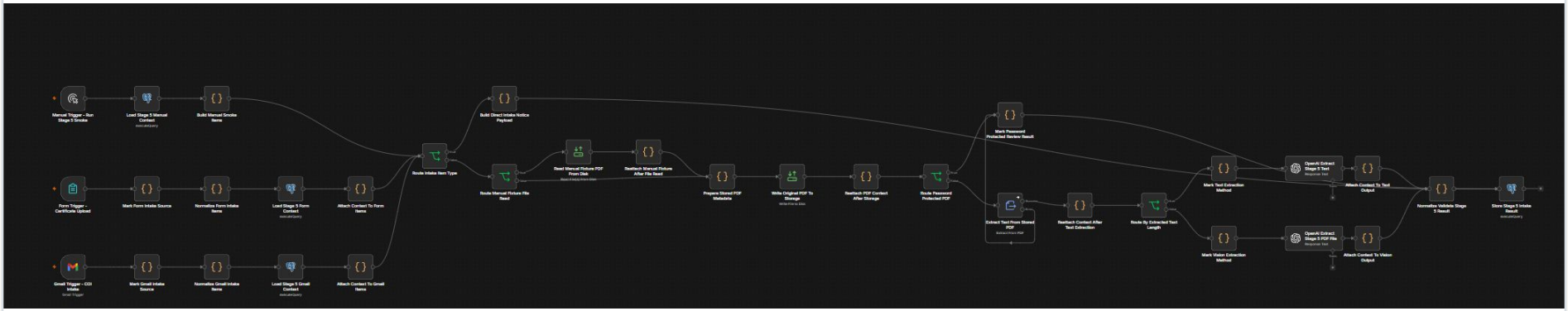
Reads Postgres and renders a vendor status register

Design boundary

This is a local demonstration built with synthetic vendors and certificates. It has not been deployed for a client or certified for production use.

Three entry paths, one intake pipeline

Direct n8n editor capture. All 35 functional nodes are visible in this overview; the next two pages enlarge the stages for reading.



What it does

- Accepts a Gmail PDF attachment, browser upload, or controlled manual fixture run.
- Normalizes the source, stores the original PDF, and keeps a source reference.
- Extracts certificate facts, runs deterministic checks, then persists the result in Postgres.

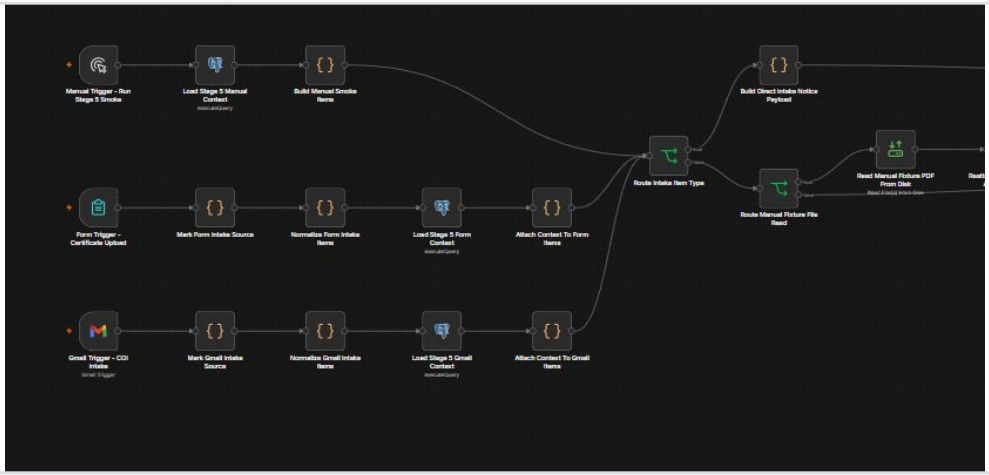
Why the flow is split

The same validation path follows each source. This keeps upload and email behavior aligned and makes fixture checks repeatable.

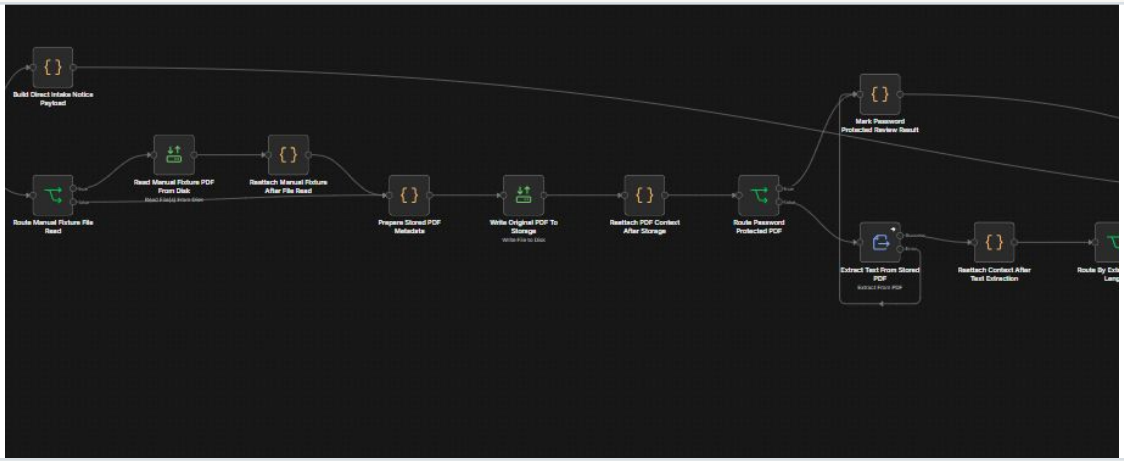
Sources and original-file handling

Enlarged views of the same canvas. These are direct n8n screenshots, cropped for legibility.

A / SOURCE BRANCHES



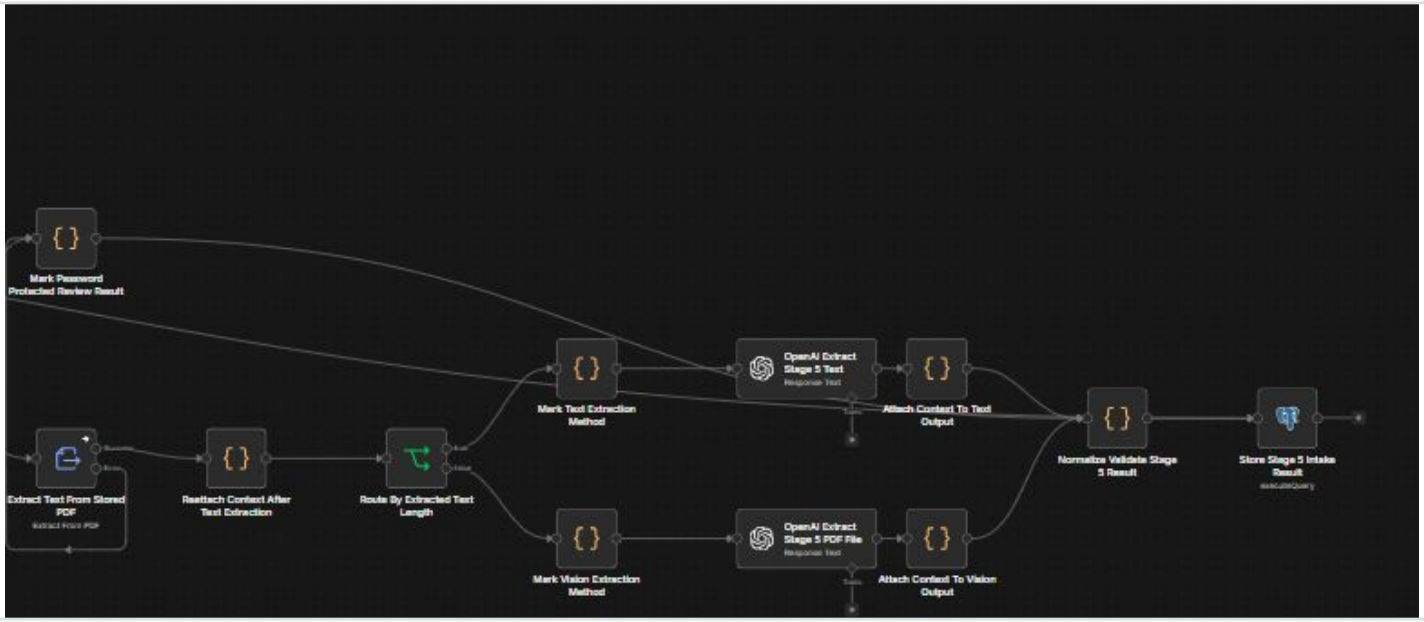
B / PDF STORAGE AND EXTRACTION BRANCH



The original is saved before parsing; a low-text PDF can take the visual extraction path.

Structured facts, fixed rules

The model transcribes document facts into a schema. Project code then applies repeatable flags and storage decisions.



Examples of checks

- Wrong holder or coverage below the minimum is flagged for review.
- A non-certificate attachment creates a notice without filing a certificate.
- Renewal history can supersede an older record and surface a coverage gap.

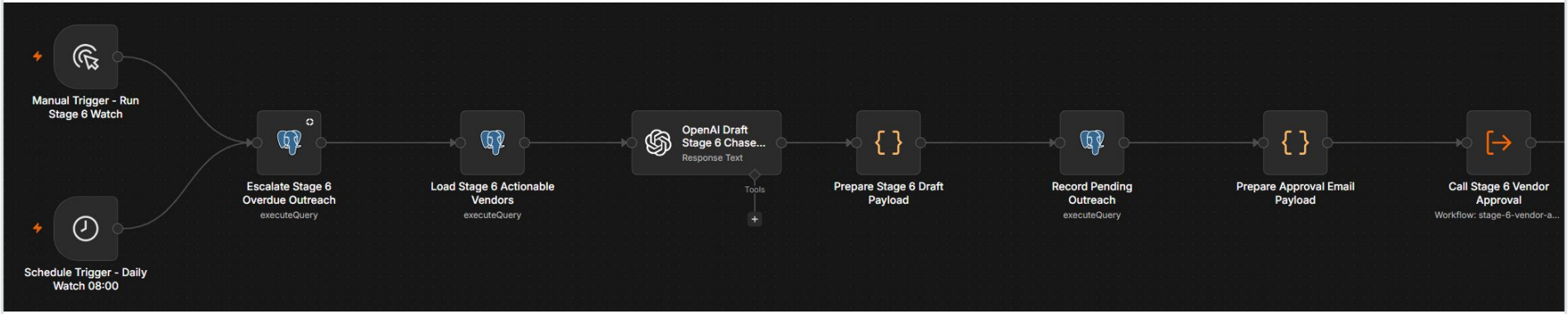
Human interpretation remains required

A filed certificate and an automation flag are operational records; they are not a legal determination that coverage is adequate.

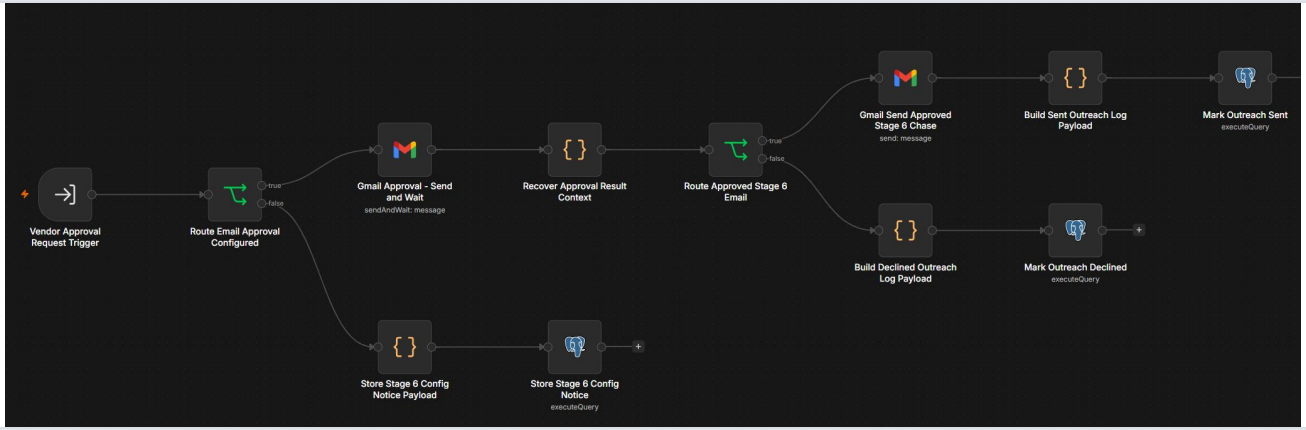
Daily watch with a human send gate

The watch selects actionable vendors. Each approval request then runs separately so every vendor gets an independent pause and decision.

DAILY WATCH AND CHASE / 9 functional nodes



VENDOR APPROVAL REQUEST / 12 functional nodes

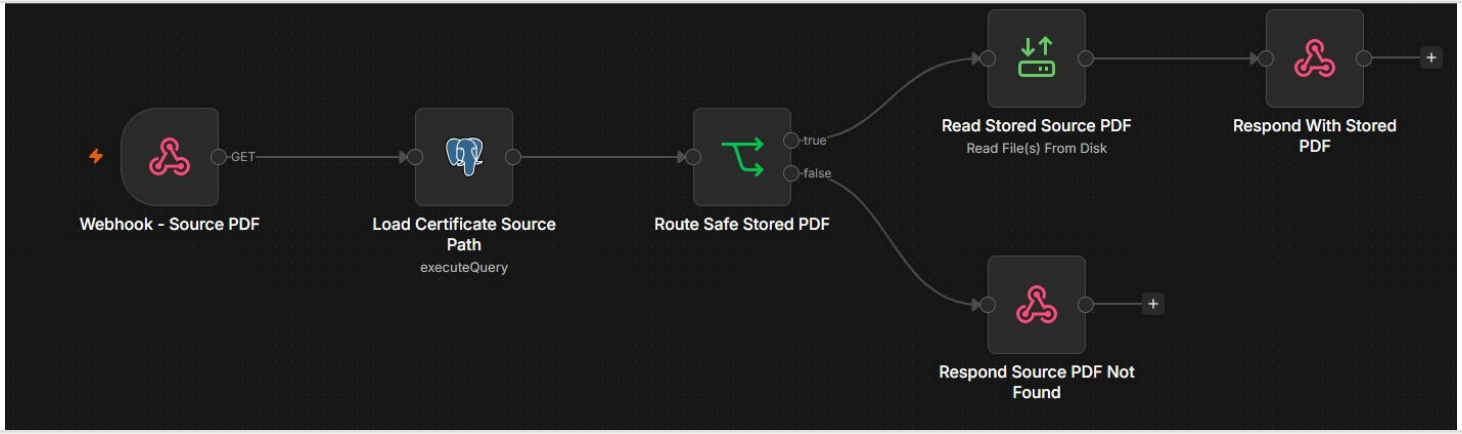


Postgres suppression and escalation rules govern who is due. Gmail approval decides whether each prepared message is sent or declined; the decision is logged.

Source access and reply capture

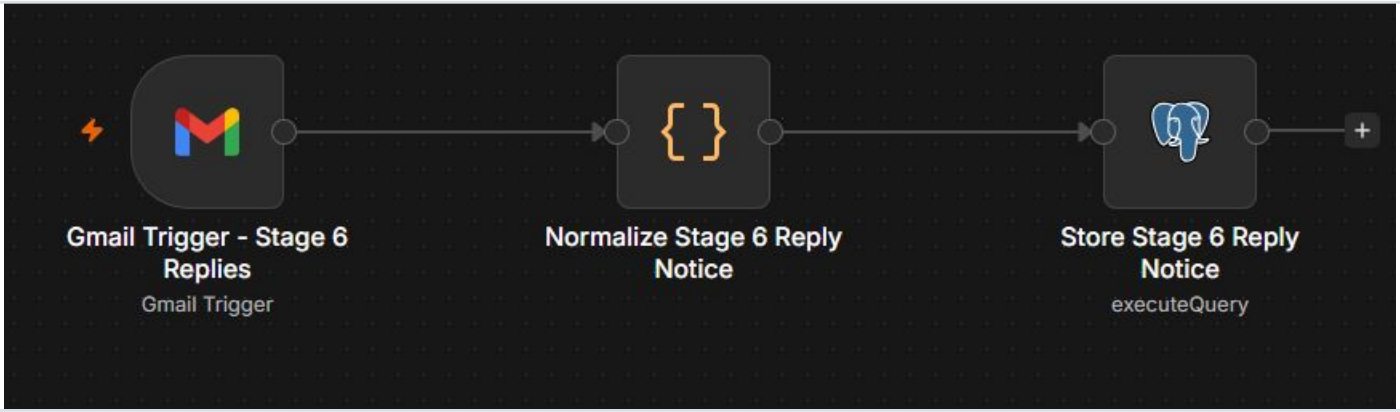
Small workflows handle two practical edge cases without making the intake canvas harder to follow.

SOURCE DOCUMENT ROUTE / 6 functional nodes



A dashboard link looks up the certificate's stored file and returns the original PDF or a not-found response.

NON-PDF REPLY NOTICE / 3 functional nodes

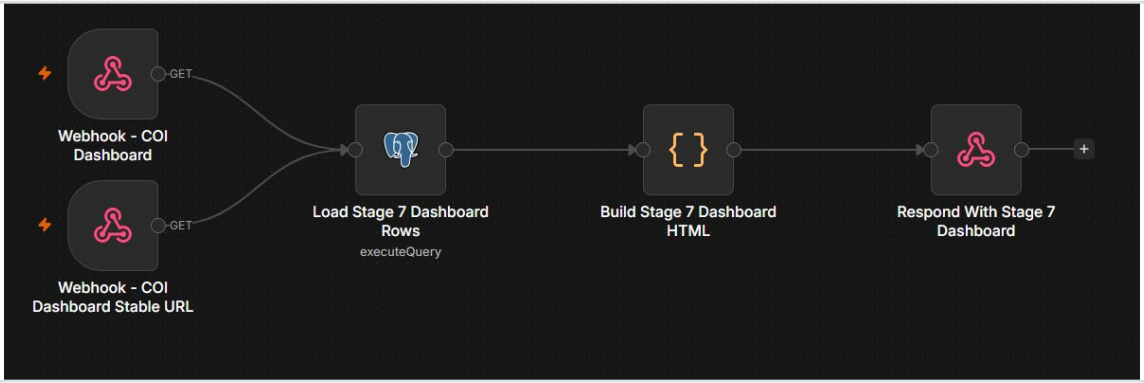


Replies without a PDF become notices for a manager; portal links are not fetched automatically.

A register backed by workflow state

The webhook reads the Postgres view and renders an HTML board with vendor status, review flags and source-document links.

DASHBOARD WORKFLOW / 5 functional nodes



RENDERED REGISTER / illustrative synthetic rows

Harbourline Property Management LLC

Vendor Insurance Register

Auto-refreshes every 5 seconds

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The source-document links were exercised in the Stage 8 fixture verification.

EVIDENCE AND SCOPE

Tested with synthetic scenarios

The repository's repeatable Stage 8 verifier passed its fixture suite and dashboard checks on the local stack.

Intake	Valid, expired, scanned, password-protected and non-certificate PDFs
Validation	Wrong holder, low limits, missing GL, inverted dates and duplicate vendor
History	Two attachments, renewal supersession and a 14-day coverage gap
Dashboard	18 seeded vendors, source links, gap count and status views
Approval	Separate per-vendor waits, sends and decline logging verified in a later live test

Scope of this sample

Local demo with synthetic documents and a test mailbox. Client hosting, access controls, alerts, retries, backups, data retention and operating terms still need implementation and testing for production use.

Evidence: docs/stage-8-fixture-results.md and workflow exports in the project repository.